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Batch 1

ROLL NO: 2

Assignment 2

Implement Simple and Multiple Linear Regression to predict continuous variables. a. Perform data preprocessing (handle missing values, feature scaling). b. Fit a Simple Linear Regression model on a dataset (e.g., predicting house prices). c. Extend to Multiple Linear Regression with multiple features. d. Evaluate models using MSE, RMSE, and R² Score. e. Visualize the regression line and predictions.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, r2_score
```

```
# Load dataset
data = pd.read_csv("/content/50_Startups.csv")

# Display first few rows
data.head()
```

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

Next steps: Generate code with data New interactive sheet

```
data = data.fillna(data.mean(numeric_only=True))
```

```
data = pd.get_dummies(data, drop_first=True)
```

```
scaler = StandardScaler()

X = data.drop("Profit", axis=1)
y = data["Profit"]
X_scaled = scaler.fit_transform(X)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)
```

Simple Linear Regression

```
X_simple = data[["R&D Spend"]]


X_simple_scaled = scaler.fit_transform(X_simple)

X_train_s, X_test_s, y_train_s, y_test_s = train_test_split(
    X_simple_scaled, y, test_size=0.2, random_state=42
)
```

```
model_simple = LinearRegression()
model_simple.fit(X_train_s, y_train_s)
```

```
▾ LinearRegression
LinearRegression()
```

```
y_pred_simple = model_simple.predict(X_test_s)
```

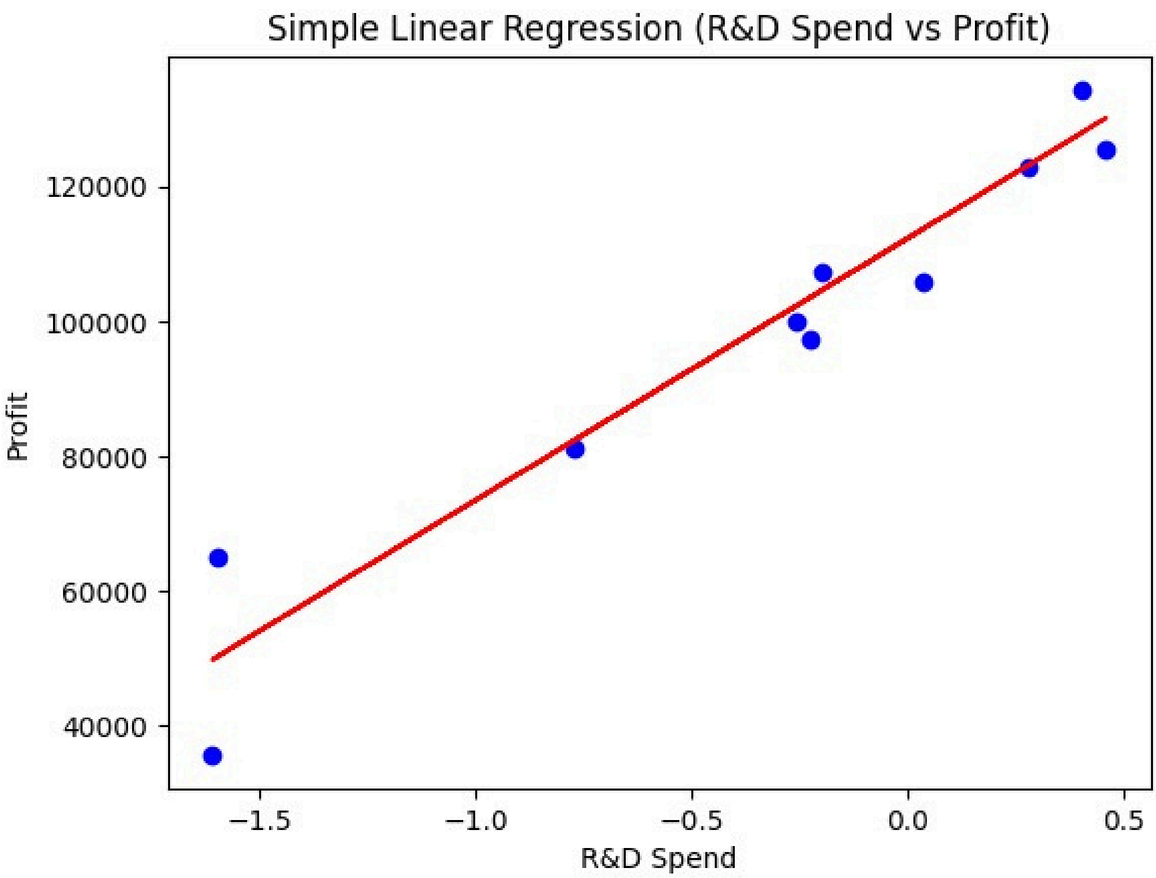
```
mse_s = mean_squared_error(y_test_s, y_pred_simple)
rmse_s = np.sqrt(mse_s)
r2_s = r2_score(y_test_s, y_pred_simple)

print("Simple Linear Regression:")
print("MSE:", mse_s)
```

```
print("RMSE:", rmse_s)
print("R2:", r2_s)
```

```
Simple Linear Regression:
MSE: 59510962.80787997
RMSE: 7714.334890830185
R2: 0.9265108109341951
```

```
plt.scatter(X_test_s, y_test_s, color="blue")
plt.plot(X_test_s, y_pred_simple, color="red")
plt.xlabel("R&D Spend")
plt.ylabel("Profit")
plt.title("Simple Linear Regression (R&D Spend vs Profit)")
plt.show()
```




```
model_multi = LinearRegression()
model_multi.fit(X_train, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

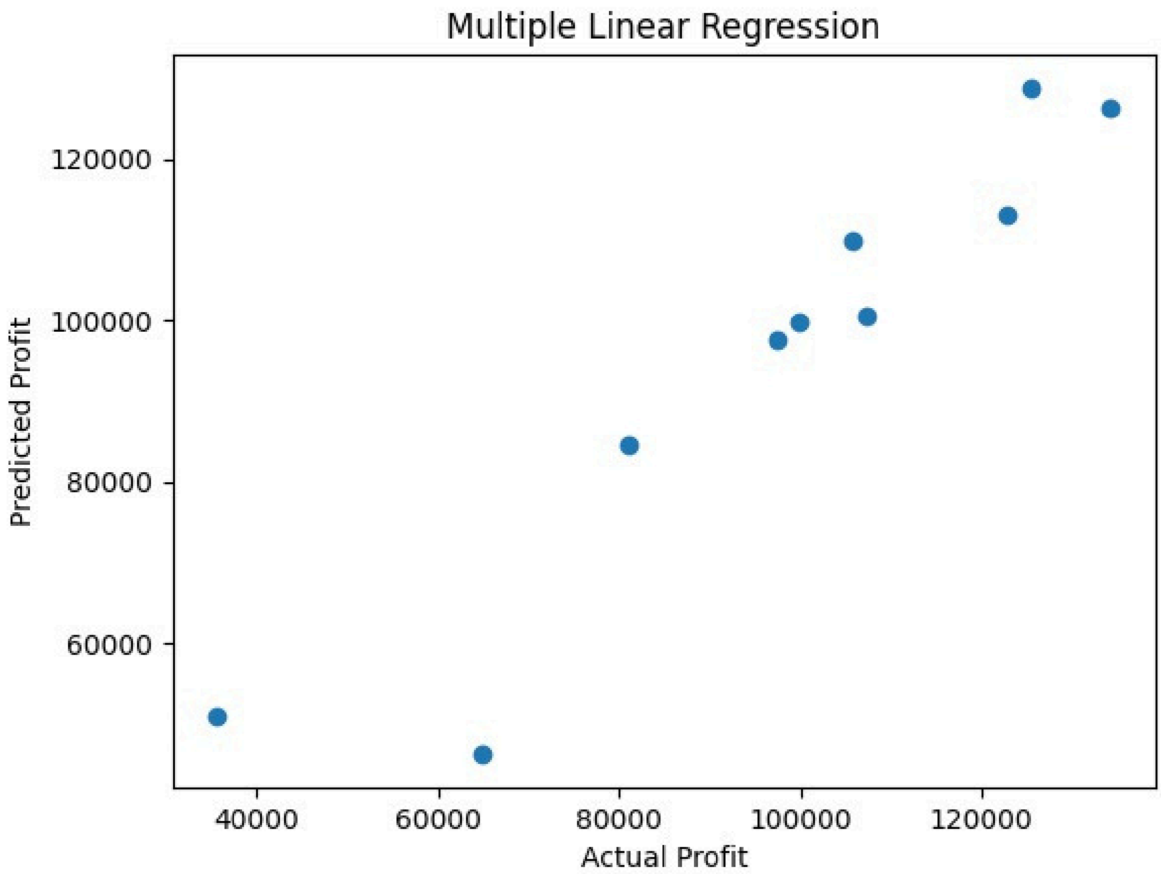
```
y_pred_multi = model_multi.predict(X_test)
```

```
mse_m = mean_squared_error(y_test, y_pred_multi)
rmse_m = np.sqrt(mse_m)
r2_m = r2_score(y_test, y_pred_multi)

print("\nMultiple Linear Regression:")
print("MSE:", mse_m)
print("RMSE:", rmse_m)
print("R2:", r2_m)
```

```
Multiple Linear Regression:
MSE: 82010363.04501365
RMSE: 9055.957323497812
R2: 0.8987266414319837
```

```
plt.scatter(y_test, y_pred_multi)
plt.xlabel("Actual Profit")
plt.ylabel("Predicted Profit")
plt.title("Multiple Linear Regression")
plt.show()
```



Conclusion

The 50_Startups dataset was used to predict startup profit.

Data preprocessing included missing value handling, encoding categorical data, and feature scaling.

Simple Linear Regression analyzed the effect of R&D spend on profit.

Multiple Linear Regression used all relevant expenditures and state information.

Performance was evaluated using MSE, RMSE, and R^2 score.

Multiple Linear Regression achieved better accuracy due to multiple contributing factors.